

AI-Powered Customer Support Ticket Triage

SaaS Delivery Implementation Plan

Professional implementation plan for a mock enterprise customer experience workflow that simulates AI-assisted ticket classification, prioritization, routing, demo readiness, training readiness, and delivery governance.

Prepared by

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Document Type	Implementation Plan
Project Type	Independent SaaS and AI Delivery Simulation
Mock Client	UrbanCart - E-commerce Customer Support Operations
Primary Focus	Requirements alignment, workflow delivery, demo readiness, training enablement, and handover planning

Portfolio Disclaimer: This document is an independent academic and portfolio simulation. It does not represent a live client engagement, production deployment, or proprietary implementation.

1. Executive Summary

This implementation plan defines a structured delivery approach for an AI-enabled SaaS workflow that assists a mock e-commerce client, UrbanCart, with customer support ticket triage. The simulated solution classifies incoming tickets, assigns priority, routes issues to the correct support team, and provides a documented path for demo, training, review, and handover.

The plan is designed from a delivery and implementation perspective rather than a production engineering perspective. It focuses on how business requirements are translated into product requirements, user stories, acceptance criteria, workflow rules, sample data, demo materials, and training assets. This makes the project suitable for demonstrating entry-level AI delivery, SaaS implementation awareness, documentation discipline, and cross-functional coordination.

The delivery method follows a lightweight Agile-style structure supported by clear stage gates, stakeholder checkpoints, requirement traceability, and implementation documentation. The approach is intentionally transparent and avoids overclaiming: AI behavior is represented through an explainable simulation and rule-based workflow logic rather than a trained machine learning model.

1.1 Implementation Objectives

- Translate the mock client problem into a clear implementation scope and delivery plan.
- Define how customer support tickets are classified, prioritized, and routed through a SaaS-style workflow.
- Create professional delivery artifacts including BRD, PRD, user stories, sample data, demo script, training guide, and handover notes.
- Demonstrate understanding of client-facing implementation, cross-functional coordination, documentation, demo preparation, and training enablement.
- Establish quality controls so the project can be reviewed confidently in a resume, GitHub portfolio, or interview setting.

1.2 Delivery Outcome

At completion, the portfolio project should present a complete mini-delivery package: problem context, requirements, workflow design, sample dataset, demo script, training guide, and resume-ready evidence of SaaS and AI delivery exposure.

2. Project Context and Delivery Rationale

UrbanCart is a mock e-commerce organization receiving high volumes of customer support tickets related to payments, refunds, delayed deliveries, cancellations, damaged products, login issues, and general queries. Manual triage creates inconsistent categorization, slower first response, missed escalations, and poor operational visibility for support managers.

The simulated SaaS workflow addresses this by introducing an AI-assisted triage layer that reads ticket text, determines the best-fit issue category, assigns urgency, and routes the case to the appropriate team. From an implementation perspective, the project focuses on how a delivery team would define requirements, document the workflow, prepare acceptance criteria, plan a demo, and support user adoption.

2.1 Delivery Principles

Principle	Application in This Project
Client-first clarity	Business pain points are documented before technical solution details are finalized.
Explainable workflow	Ticket classification and priority rules are documented so stakeholders can understand why an output is produced.
Controlled scope	The project remains a simulation and does not claim real AI model training or live SaaS deployment.
Traceable delivery	BRD items connect to PRD items, user stories, acceptance criteria, and demo scenarios.
Adoption readiness	Training guide and demo script are prepared to show how support users would understand the workflow.

3. Scope Definition

The scope is intentionally defined to demonstrate implementation planning and delivery documentation. It includes the artifacts and workflow logic required to explain a SaaS-style AI delivery project while excluding production-level engineering claims.

3.1 In Scope

- Requirement understanding and business problem framing for the mock client.
- Ticket category definition for refund, delivery delay, payment issue, damaged product, cancellation, login issue, and general query.
- Priority assignment logic for high, medium, and low urgency tickets.
- Team routing rules for support operations.
- Sample ticket dataset for demonstrating workflow behavior.
- BRD, PRD, user stories, acceptance criteria, implementation plan, demo script, and training guide.
- GitHub repository structure for portfolio presentation and interview discussion.

3.2 Out of Scope

- Production SaaS deployment or live platform configuration.
- Training, fine-tuning, or benchmarking of a real machine learning model.
- Processing of real customer data, personal data, or payment information.
- Integration with CRM, payment gateway, logistics systems, or identity providers.
- Enterprise-grade security architecture, performance engineering, and production monitoring.
- Commercial service-level agreement commitments or live support operations.

4. Stakeholder Model and Delivery Responsibilities

A strong implementation plan clarifies stakeholder roles even in a simulation. The following model shows how a real delivery team might collaborate across business, product, technology, support, and customer success functions.

Stakeholder	Primary Responsibility	Expected Contribution
Support Agents	Use ticket category, priority, and routing output.	Validate whether categories and priorities are practical for day-to-day support work.
Support Manager	Monitor volumes, escalations, and operational effectiveness.	Confirm success metrics, priority rules, and escalation expectations.

Stakeholder	Primary Responsibility	Expected Contribution
Implementation Team	Coordinate requirements, documentation, and delivery activities.	Maintain BRD, PRD, user stories, action items, demo readiness, and handover checklist.
Product Team	Review feature scope and workflow behavior.	Validate functional requirements and identify product assumptions or gaps.
Technology Team	Translate requirements into configurable or buildable workflow logic.	Review classification rules, routing logic, data assumptions, and technical feasibility.
Customer Success Team	Support adoption, demo preparation, and training.	Ensure users understand the workflow and expected business value.

5. Workstream Breakdown

The implementation is divided into six workstreams so that documentation, workflow design, sample data, demo readiness, quality review, and handover remain organized.

Workstream	Purpose	Key Output
Requirements	Capture business goals, scope, stakeholders, and success metrics.	BRD and open questions register.
Product Definition	Convert business needs into product behavior and acceptance criteria.	PRD and user stories.
Workflow Design	Define classification, prioritization, and routing logic.	Workflow diagram and routing matrix.
Data Simulation	Create safe sample tickets for demonstration and validation.	Customer tickets CSV.
Demo and Training	Prepare client-facing explanation and user enablement material.	Demo script and training guide.
Review and Handover	Check consistency, polish repository, and prepare interview narrative.	Final GitHub project and resume bullets.

6. Delivery Lifecycle

The delivery lifecycle uses a phased implementation model. Each phase produces a tangible artifact and includes a quality gate to ensure that the work is ready before moving forward.

Phase	Focus	Activities	Exit Criteria
Phase 1 - Requirement Understanding	Define the business problem and client context.	Identify problem, stakeholders, objectives, ticket categories, priority logic, and routing needs.	Business problem, scope, and initial assumptions are documented.
Phase 2 - Product Requirement Documentation	Convert business goals into product-ready requirements.	Prepare PRD, functional requirements, non-functional requirements, user stories, and acceptance criteria.	PRD and user stories are complete and aligned with BRD.
Phase 3 - Workflow Design	Design the ticket triage workflow.	Map ticket categories, priority conditions, routing teams, sample data, and workflow diagram.	Workflow is explainable, reviewable, and connected to sample tickets.
Phase 4 - Demo Preparation	Prepare a client-facing demonstration.	Select sample tickets, write demo narrative, explain outputs, and document support team usage.	Demo can be delivered clearly in 3 to 5 minutes.
Phase 5 - Review and Handover	Prepare final portfolio package.	Review consistency, polish GitHub repository, prepare resume bullets, and define interview talking points.	Final project is organized, honest, and ready for resume submission.

7. Detailed Phase Plan

7.1 Phase 1 - Requirement Understanding

The first phase establishes the business foundation for the implementation. The delivery owner reviews the mock client scenario, defines operational challenges, identifies stakeholders, and confirms the problem statement. This phase ensures that the project starts with a business need rather than with a tool-first solution.

- Document current support pain points such as delayed triage, inconsistent routing, and missed urgent cases.
- Define ticket categories and priority conditions from the client support perspective.
- Capture assumptions, risks, and questions that would require client confirmation in a real implementation.
- Produce the BRD and initial open questions register.

7.2 Phase 2 - Product Requirement Documentation

This phase translates the business problem into product-level behavior. The PRD defines the feature objective, user roles, system behavior, functional requirements, non-functional requirements, user stories, and acceptance criteria.

- Define the expected input and output of the triage workflow.
- Map business requirements to functional requirements.
- Write user stories for support agents, support managers, implementation team members, and customer success roles.
- Define acceptance criteria that can be demonstrated using sample tickets.

7.3 Phase 3 - Workflow Design

This phase creates the operational logic behind the simulation. The workflow describes how a ticket enters the SaaS platform, how it is classified, how priority is assigned, and how the routing team is selected.

- Create category-to-team mapping.
- Define priority rules based on issue type, urgency, customer sentiment, and escalation language.
- Prepare sample tickets to validate expected output.
- Create a workflow diagram to make the process visually explainable.

7.4 Phase 4 - Demo Preparation

The demo phase prepares a structured walkthrough for a recruiter, interviewer, or mock client. The demo should show the problem, workflow, example input, expected output, business value, limitations, and next steps.

- Select 3 to 5 representative sample tickets for walkthrough.
- Explain the classification, priority, and routing output for each sample ticket.
- Highlight that the project demonstrates delivery thinking, not production AI engineering.
- Prepare a 60-second and 3-minute version of the explanation.

7.5 Phase 5 - Review and Handover

The final phase ensures the project is polished, consistent, and ready to be shared. The handover package should be easy to navigate and should make the candidate appear organized, honest, and delivery-focused.

- Review all documents for terminology consistency and spelling accuracy.
- Confirm that the README links to or references all supporting documents.
- Add resume bullets that describe the project as a simulation, not real client experience.
- Prepare interview talking points that connect the project to AI delivery and SaaS implementation roles.

8. Requirements Traceability and Acceptance Strategy

Requirement traceability ensures that each business requirement can be connected to a product behavior, artifact, or demo scenario. This is especially important in implementation roles because it shows that requirements were not only written but also converted into reviewable outputs.

Business Requirement	Linked Product Behavior	Evidence Artifact
Classify incoming tickets into predefined categories.	Ticket category output appears for every sample ticket.	PRD, User Stories, Sample Data, Demo Script
Assign priority based on issue type and urgency.	High, Medium, or Low priority is assigned based on documented logic.	PRD, Priority Matrix, Demo Script
Route tickets to the correct team.	Each category maps to a routing team.	Routing Matrix, Workflow Diagram, Training Guide
Flag urgent cases for faster handling.	High-priority cases are identified and explained during demo.	Demo Script, Training Guide
Support manual review for unclear tickets.	Unmatched or ambiguous tickets can be marked for manual review.	PRD, User Stories, Risk Controls

8.1 Acceptance Criteria Summary

- Every sample ticket should have a category, priority, and routing team.
- Each category should have a documented support team mapping.
- High-priority scenarios should be explainable through business rules.
- Demo script should be understandable to a non-technical audience.
- Training guide should help a support agent or manager understand how to use the output.
- Repository should clearly show that this is a portfolio simulation.

9. Governance, Communication, and Review Cadence

Although this is an independent portfolio project, the implementation plan uses professional governance concepts to demonstrate delivery maturity. In a real implementation, governance would help manage scope, decisions, risks, and stakeholder alignment.

Governance Area	Professional Practice Used in This Project
Scope Control	In-scope and out-of-scope items are explicitly documented to avoid overclaiming.
Decision Tracking	Assumptions and open questions are recorded for review.
Stakeholder Alignment	Stakeholder roles and responsibilities are defined in the delivery model.
Quality Review	Documents are checked for consistency across BRD, PRD, user stories, demo, and training guide.
Portfolio Readiness	README, document names, and resume bullets are aligned to the JD without claiming live implementation experience.

9.1 Communication Plan

Audience	Communication Need	Recommended Artifact
Recruiter	Quick understanding of project relevance.	README and resume bullets.
Hiring Manager	Evidence of delivery thinking and documentation quality.	Implementation Plan, BRD, PRD, Demo Script.
Product/Implementation Interviewer	Understanding of requirements, user stories, scope, and acceptance criteria.	PRD, User Stories, Traceability Matrix.
Technical Interviewer	Understanding of workflow logic and sample data.	Workflow Diagram, Sample Data, Optional Script.

10. Risk Management

Risk management is included to show how a delivery associate or implementation trainee can think beyond task completion. The objective is to identify risks early and define practical mitigations.

Risk	Potential Impact	Mitigation
Overclaiming AI capability	Project may appear misleading if described as a real AI model.	Use terms such as AI-enabled simulation, rule-based logic, and portfolio project.
Unclear scope	Reviewer may not understand what was actually implemented.	Keep scope and out-of-scope sections clear in README and documents.
Inconsistent documents	BRD, PRD, and demo may describe different logic.	Review category names, priority rules, and team routing across all files.
Weak business connection	Project may look technical but not relevant to AI delivery role.	Emphasize client requirements, workflow, demo, training, and handover.
Poor repository navigation	Recruiter may not find the important files quickly.	Use clear folders and a strong README with document list.

11. Quality Gates and Completion Checklist

The following quality gates ensure that the project is ready for submission and interview discussion.

Quality Gate	Completion Standard
Documentation Quality	BRD, PRD, implementation plan, demo script, and training guide are complete, professional, and internally consistent.
Requirement Alignment	Business requirements connect to product requirements, user stories, and acceptance criteria.
Workflow Readiness	Classification, priority, and routing logic are easy to explain using sample tickets.
Demo Readiness	Candidate can explain the project in 60 seconds and expand into a detailed walkthrough.
Resume Readiness	Project bullets accurately describe a simulation and include relevant JD keywords.
GitHub Readiness	Repository is public, organized, and includes a clear disclaimer.

11.1 Final Checklist

- README explains the project purpose, mock client, problem, solution, artifacts, and disclaimer.
- BRD captures business problem, objectives, stakeholders, scope, requirements, metrics, assumptions, and risks.
- PRD captures functional and non-functional requirements, user roles, workflow behavior, and open questions.

- User stories include acceptance criteria for support agents, managers, implementation team, and customer success roles.
- Sample data includes realistic ticket messages with expected category, priority, and routing team.
- Demo script explains the project clearly without claiming production AI implementation.
- Training guide explains how support users would interpret categories, priorities, and routing outputs.
- Resume project section includes the GitHub link and uses honest, ATS-friendly wording.

12. Handover Package and Interview Positioning

The final handover package should help reviewers quickly understand the project and should help Jagriti Banerjee explain the work confidently during screening or interview conversations.

Artifact	Purpose
README.md	Front-page project explanation and navigation guide.
BRD.md / BRD PDF	Business problem, scope, stakeholders, requirements, and risks.
PRD.md / PRD PDF	Product behavior, functional requirements, workflow rules, and acceptance criteria.
User-Stories.md	Role-based requirements and acceptance criteria.
Implementation-Plan.md / PDF	Delivery approach, phases, governance, risks, quality gates, and handover plan.
Demo-Script.md / PDF	Client-facing walkthrough and interview explanation.
Training-Guide.md	End-user support guide for agents and managers.
customer_tickets.csv	Sample ticket data used to demonstrate classification and routing behavior.

12.1 Interview Positioning Statement

This project can be positioned as follows:

"I built this independent SaaS delivery simulation to understand how AI-enabled customer support workflows are delivered from a business and implementation perspective. I created the BRD, PRD, user stories, implementation plan, demo script, training guide, sample data, and workflow logic. The goal was not to claim production AI model development, but to demonstrate requirement documentation, client-facing delivery thinking, cross-functional coordination, and SaaS implementation readiness."

13. Professional Summary of Delivery Value

This implementation plan demonstrates structured thinking across requirement capture, product documentation, workflow design, demo preparation, training enablement, risk management, quality control, and project handover. It connects technical project discipline with business-facing AI delivery expectations and provides a clear portfolio artifact for entry-level SaaS implementation and AI delivery roles.

For a resume or interview, the strongest value of this project is not the complexity of the code. The strongest value is the ability to understand a client problem, convert it into structured requirements, define workflow behavior, prepare delivery artifacts, and communicate the solution clearly to business and technical stakeholders.